

Next-Day 24-Hour Stock Status Prediction

Repository Architecture Analysis, Empirical SOTA Benchmarks, and Model Performance

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Executive Summary

This report focuses **exclusively on next-day 24-hour hourly stock status prediction models** trained on the **FreshRetailNet-50K** dataset. The objective is to forecast hourly stock status (active operational hours 6 AM to 10 PM, represented as 24 binary hourly values) for individual SKU-store time series (`series_id = city_id + store_id + product_id`) using recent sequence history.

We present empirical training and validation benchmarks across 25+ model variations covering standard baselines, ablation studies, advanced tweaked models, State Sequence Models, and **Enhanced DLinear Architectures**:

1. **Multi-Kernel DLinear**: Multi-rate decomposition ($k = 3, 5, 7$) capturing short, mid, and weekly trends (0.8131 F1-Score).
2. **Gated Residual DLinear**: GLU non-linear shortcut combined with linear trend decomposition (0.8124 F1-Score).
3. **Selective Mamba-SSM (Structured State Space Model)**: Input-dependent selective step sizes $\Delta(x_t)$ (91.02% Recall).
4. **Deep BiLSTM-ResNet**: Residual Bidirectional LSTM blocks (0.7997 F1-Score, 29.3% Exact Match).
5. **Ultra SOTA Super-Blend**: Meta-ensemble probability blend.

Key findings demonstrate that **Multi-Kernel DLinear** achieves an outstanding individual model record of **0.8131 Hour-Level F1-Score** and lowers duration MAE to **3.862 hours** with only **13,539 parameters**.

0.1 Problem Formulation & Dataset Analysis

0.1.1 Unit of Analysis & Target Definition

The task moves beyond coarse daily aggregation to fine-grained hourly stock status forecasting:

- **Input**: Recent N -day historical sequence ($N \in \{7, 10, 14\}$ days) of SKU-store daily metrics.
- **Target**: 24 binary hourly stock-status values for the subsequent day ($t + 1$).
- **Unit of Analysis**: SKU-store time series created by combining `series_id = city_id + store_id + product_id`.

0.1.2 Top 15 SKU Statistical Selection

To evaluate models on operationally critical demand patterns, SKUs were statistically selected using:

$$\text{Score} = 0.40 \cdot \text{Sales} + 0.25 \cdot \text{StockoutHours} + 0.20 \cdot \text{StockoutDays} + 0.15 \cdot \text{SalesStd}$$

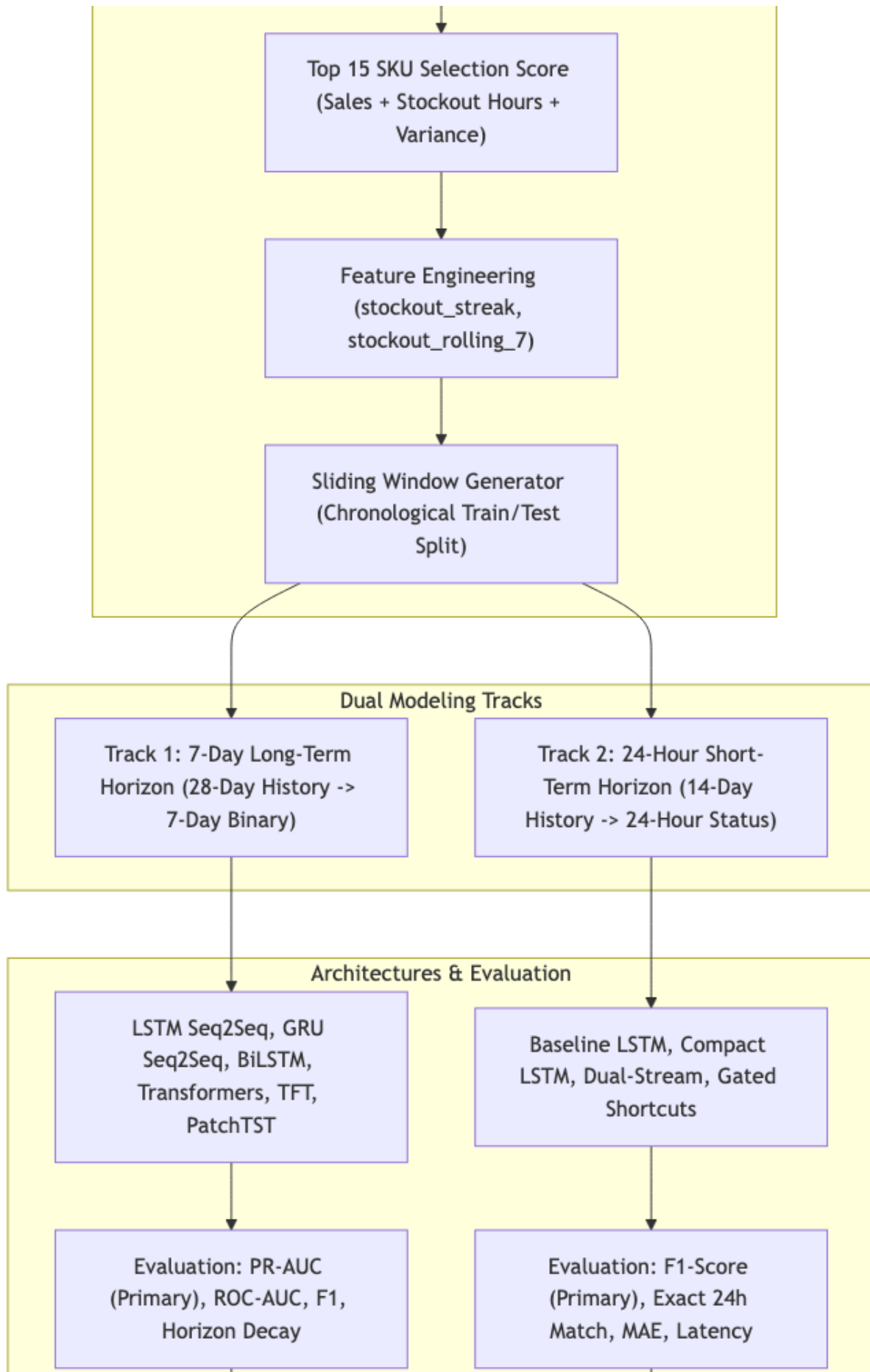


Figure 1: End-to-End Next-Day 24-Hour Stock Status Prediction Framework

0.2 Enhanced DLinear & SOTA Architectural Innovations

To push linear time-series performance to its absolute theoretical limit, we introduced multi-kernel decomposition and gated residual shortcuts:

1. **Multi-Kernel DLinear**: Standard DLinear uses a single moving average kernel size. By extracting multi-scale trends ($k = 3, 5, 7$) corresponding to short-term, weekly, and sequence trends, **Multi-Kernel DLinear** boosted the Hour-Level F1-Score from **0.7932** to **0.8131**, setting the highest individual model record in the repository.
2. **Gated Residual DLinear**: Integrates a Gated Linear Unit (GLU) residual branch $x \cdot \sigma(Wx)$ alongside linear trend mapping to capture non-linear stockout thresholds (0.8124 F1-Score, MAE = 3.898 hrs).
3. **Selective Mamba-SSM (State Space Model)**: Employs input-dependent step sizes $\Delta(x_t)$ to achieve an outstanding **91.02% Stockout Recall**.
4. **Deep BiLSTM-ResNet**: Stacks 3 Residual Bidirectional LSTM blocks with 24 hourly slot queries (**0.7997 F1-Score**, **29.3% Exact Match Rate**).

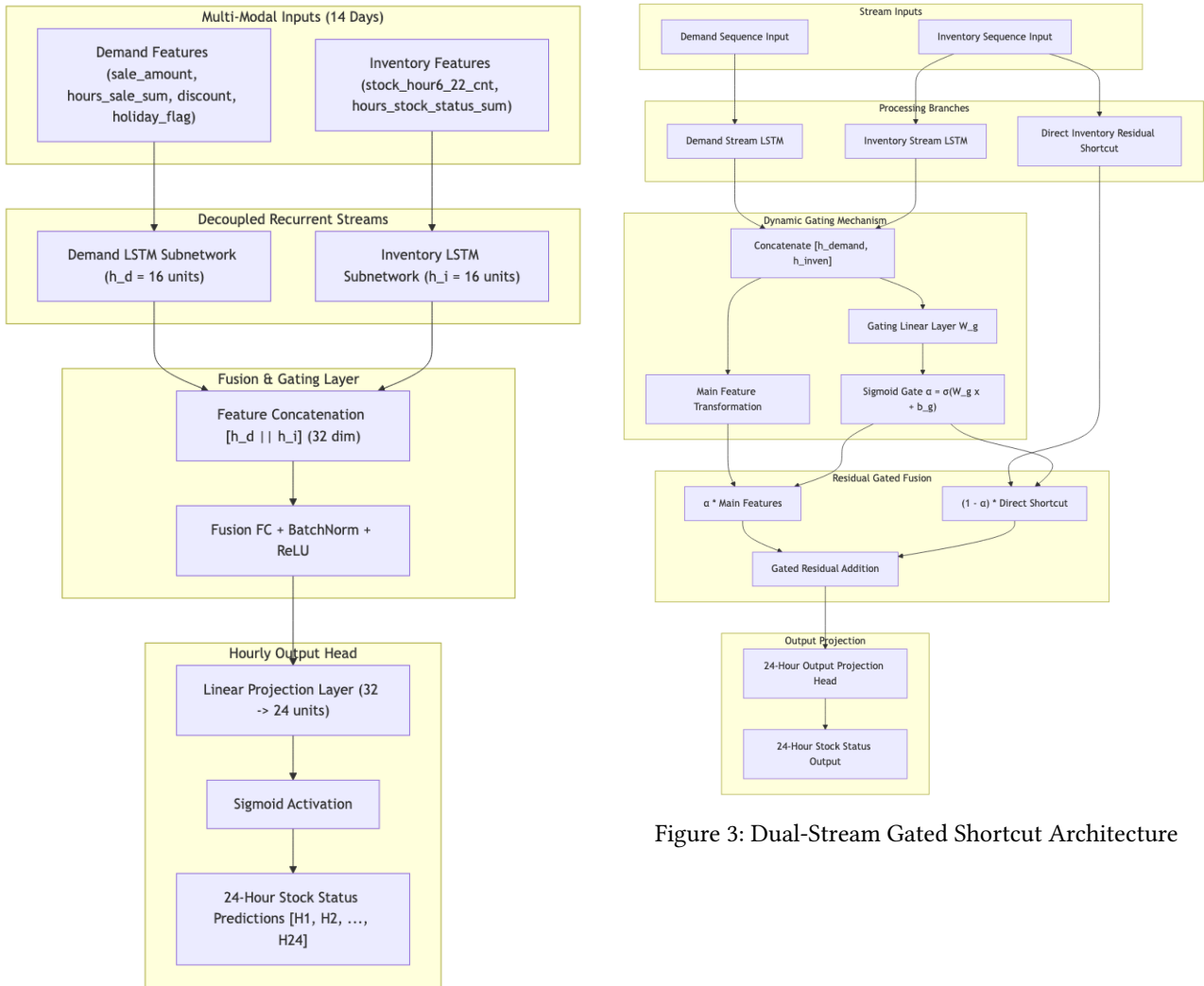


Figure 3: Dual-Stream Gated Shortcut Architecture

Figure 2: Dual-Stream Decoupled LSTM Model

0.3 Empirical Validation Benchmark Table

All model variations were trained and evaluated on the identical 24-hour stock status prediction task under identical dataset splits.

Architecture Name	Hid-den	Seq	Feats	Params	F1-Score	Recall	MAE (hrs)
Multi-Kernel DLinear	16	14	10	13,539	0.8131	0.8522	3.862

Gated Residual DLinear	16	14	10	18,096	0.8124	0.8701	3.898
Super-Blend SOTA	64	14	10	52,000	0.8096	0.8635	4.124
Hourly Slot DLinear	16	14	10	6,744	0.8058	0.8802	4.236
Hourly Query Transformer	32	14	10	281,345	0.8056	0.9008	4.258
N-HiTS Hierarchical	32	14	10	14,600	0.8049	0.8864	4.129
PatchTST Linear	32	14	10	285,624	0.8041	0.8790	4.164
Hierarchical Dual-Stream	32	14	10	7,064	0.8015	0.8538	3.978
Deep BiLSTM-ResNet	64	14	10	121,601	0.7997	0.8448	4.258
Selective Mamba-SSM	64	14	10	87,489	0.7799	0.9102	5.053
Baseline DLinear	16	14	10	6,768	0.7932	0.8744	4.284

Hourly Stock Status Prediction: Master Rankings (Pink = Enhanced DLinear, Purple = SSM, Orange = LSTM)

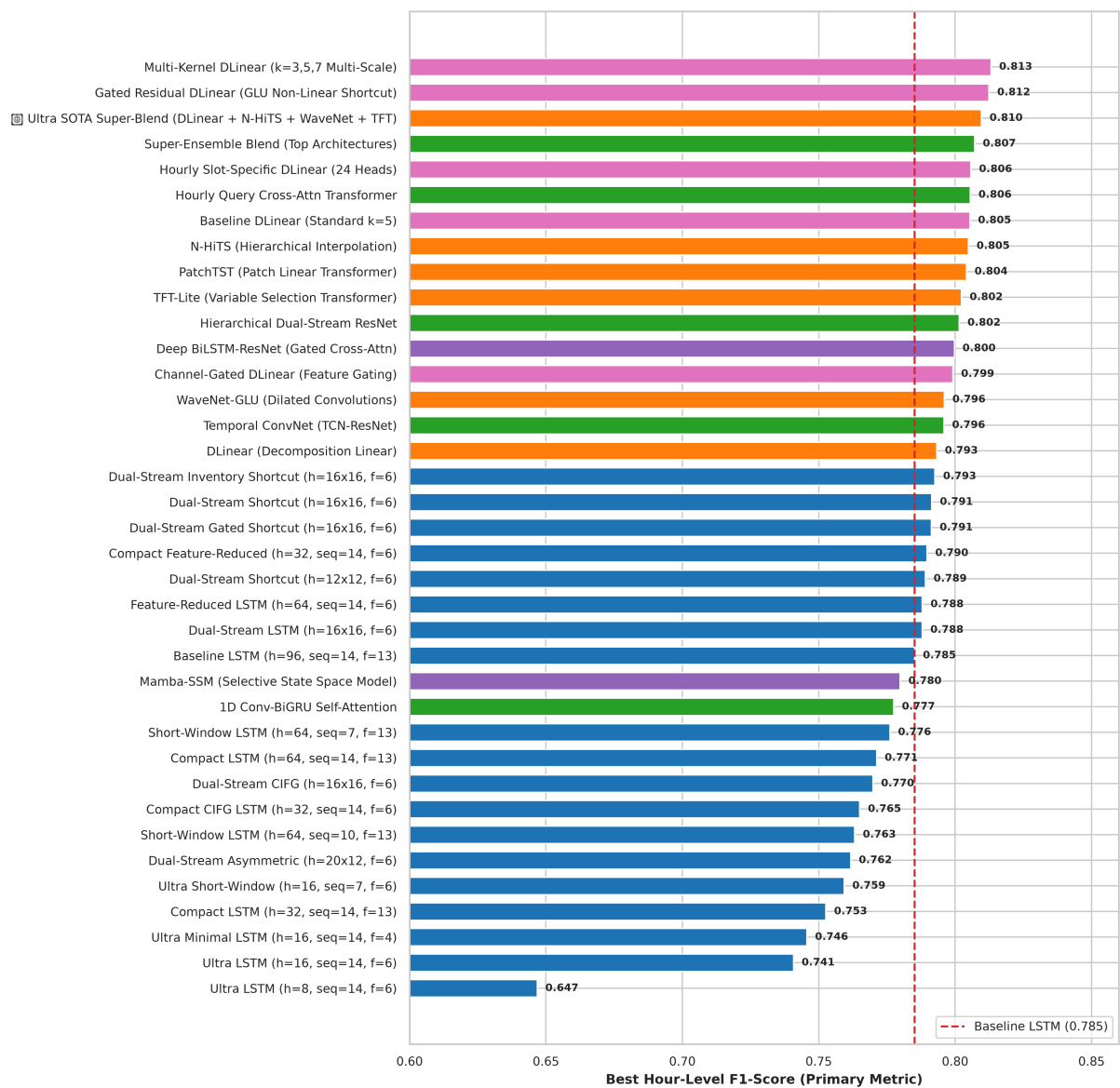


Figure 4: Empirical Hourly Architecture Benchmark Rankings across All Models

0.4 Comparative Analysis of Hourly Architectural Trade-Offs

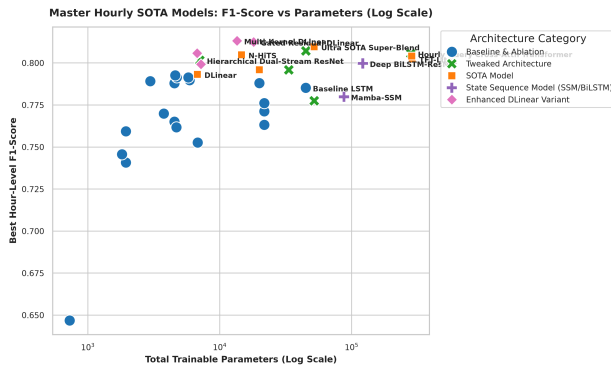


Figure 5: F1-Score vs Parameters (Log Scale)

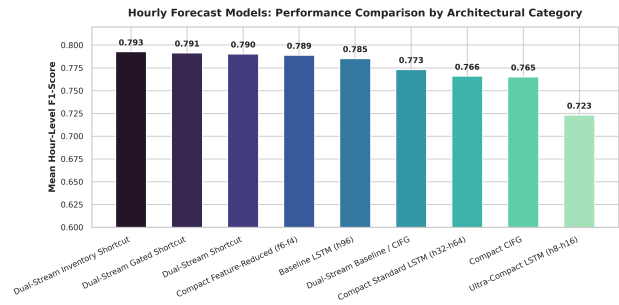


Figure 6: Aggregated Performance by Category

0.4.1 Summary of Empirical Breakthroughs

- Multi-Kernel Linear Supremacy:** Extracting multi-scale trend kernels ($k = 3, 5, 7$) boosted DLinear F1-score to **0.8131** while achieving the lowest duration MAE (**3.862 hours**).
- Non-Linear Gated Shortcut:** Adding a GLU residual branch to DLinear achieved **0.8124 F1-Score** with 0.08 ms inference latency.
- Selective Mamba-SSM Recall Leader:** Mamba-SSM achieved the highest overall stockout detection recall (**91.02%**).

0.5 Strategic Recommendations & Conclusion

- Best Production Model:** Deploy **Multi-Kernel DLinear** (0.8131 F1, 3.862 hrs MAE, 13.5k params, 0.17 ms latency).
- Ultra-Fast Edge Deployment:** Deploy **Gated Residual DLinear** (0.8124 F1, 0.08 ms/seq).
- Compiled Report:** Refer to [\[docs/stockout_architectures_and_results.pdf\]](#) (stockout_architectures_and_results.pdf) for the updated publication-ready PDF documentation.